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Optimization of the Formulation of Drilling Mud using Box – Behnken Design

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Abstract

This study examines the formulation and optimization of drilling mud utilizing locally sourced materials, specifically cassava starch, corn cob ash, and sodium hydroxide (NaOH), as economical and sustainable alternatives. The clay was evaluated, formulated, and optimized using Response Surface Methodology, specifically employing the Box-Behnken Design to enhance critical rheological parameters, including apparent viscosity and plastic viscosity. Experimental results demonstrate that these locally sourced additives enhance drilling fluid performance, providing competitive rheological characteristics comparable to conventional drilling muds. The optimization study confirmed that the developed models for apparent and plastic viscosity are highly significant, with strong predictive capabilities. ANOVA results showed that clay, ash, and starch significantly influenced apparent viscosity, with interaction effects such as clay-ash and clay-starch playing key roles. The high R^2 value (0.9737) and adequate precision (19.2750) validated the model's reliability. Similarly, plastic viscosity was significantly affected by clay and ash, with notable nonlinear interactions. The R^2 value (0.9566) and adequate precision (14.5243) confirmed model accuracy. Results obtained shows the optimum apparent viscosity of 116.431 mPa.s and plastic viscosity of 75.6167 mPa.s with the weight of the addictives being clay: 20g, Ash: 1.5g, NaOH: 2g, and starch: 5g. Overall, the findings suggest that optimizing clay and starch content enhances apparent viscosity, while clay and ash primarily influence plastic viscosity, improving drilling mud performance

1. Introduction

Drilling mud, also known as drilling fluid, plays a vital role in oil and gas well operations by cooling and lubricating the drill bit, stabilizing the wellbore, controlling formation pressures, and carrying drill cuttings to the surface [1]. Conventional mud formulations rely heavily on imported additives like bentonite and synthetic polymers, which are expensive and prone to supply disruptions. Growing interest in using locally sourced materials has led to research demonstrating Nigeria's potential to produce drilling fluids using resources such as cassava starch, corn cob ash, and sodium hydroxide (NaOH), all of which enhance the fluid's performance [2]. To meet industry requirements, optimizing these formulations is essential, and the Response Surface Methodology (RSM), specifically the Box-Behnken design, offers an effective way to fine-tune the composition.

This approach aims to create a cost-efficient and technically reliable drilling mud suited for Nigeria's oil and gas industry [3]. Nigeria's reliance on imported drilling mud additives creates economic, logistical, and environmental concerns, while the inconsistent quality of local materials challenges their use. This study focuses on the formulation and optimization of drilling mud using cassava starch, corn cob ash, and NaOH. This research provides valuable insights into the application of Response Surface Methodology (RSM) in drilling mud formulation [4]. By demonstrating the effectiveness of statistical optimization techniques, this study can serve as a reference for future research in drilling fluid development and other related fields in chemical engineering.

The adoption of locally sourced drilling mud additives has the potential to enhance the efficiency, sustainability, and economic viability of oil and gas exploration in Nigeria [5]. This study aims to optimize the formulation of drilling mud made from locally sourced cassava starch, corn cob ash, and NaOH using Response Surface Methodology (RSM) to achieve industry standards, ultimately reducing costs, import dependence, and encouraging local material use [6][1].

2. Materials and Methods

2.1.Materials

The local clay was obtained from Ovia River, Edo State, Nigeria. The Ovia River flows between latitudes 5 O 20 and 6 O 35 N and longitudes 5 O 05 and 5 O 40 E . The Clay was then dried and crushed [7].

2.2 Characterization of clay sample

Scanning Electron Microscopy (SEM) was used to analyze the microstructural properties of the clay sample [5]. The images provide insight into the morphology, porosity, and particle distribution, which are essential characteristics influencing its performance in drilling mud formulations [8].

2.3 Cassava starch

The cassava was washed, peeled, and ground into a smooth pulp, then soaked in water to release the starch, which was filtered using a fine mesh. Once the starch settled, the excess liquid was drained off, and the starch was dried completely. The dried starch was then stored securely in an airtight container for later use [9].

2.4 Corn Cob ash

The corn cobs were completely dried and then incinerated in a furnace at 600°C for two hours until they were reduced to a fine ash. Once cooled for safe handling, the ash was ground to a consistent powder. The processed corn cob ash was then sealed in an airtight container to protect its quality [10]

2.5 Formulation of drilling mud

The drilling mud was prepared using a step-by-step mixing process to ensure even distribution of all ingredients [11]. First, distilled water (350 mL) was added to a mixing cup, followed by NaOH (0.5 g) to adjust the pH. Local clay was gradually mixed in while stirring, then corn cob ash and cassava starch were added to improve filtration loss control and viscosity. Finally, the mixture was stirred for 30 minutes to achieve uniform consistency, then left to age for 10 minutes before testing [12].

2.6 Determination of Viscosity

The Baroid (Model 286) Rheometer was used to determine multipoint viscosities. It is a coaxial cylindrical rotational viscometer with fixed speeds of 3 (GEL), 100, 200, 300 and 600 RPM (revolution per minute) that are switch selectable with the RPM knob.

2.7 Determination of pH

The pH meter was calibrated using deionized water, and the mud sample was placed in a glass beaker. The pH meter probe was immersed in the sample, and once a stable pH reading appeared, it was recorded as the mud's pH value.

2.8 Optimization with BBD

The Box-Behnken Design (BBD) was used to optimize the viscosity of drilling mud by analyzing multiple factors with minimal experiments, generating 29 runs [2]. After preparing samples based on the BBD matrix and measuring viscosity, a quadratic polynomial model was developed and validated through regression analysis and ANOVA. The optimized formulation was confirmed by comparing predicted and actual viscosity values, showing a strong model fit with a high R^2 value [6].

3.0 Result and Discussion

3.1 Analysis of Clay SEM Images

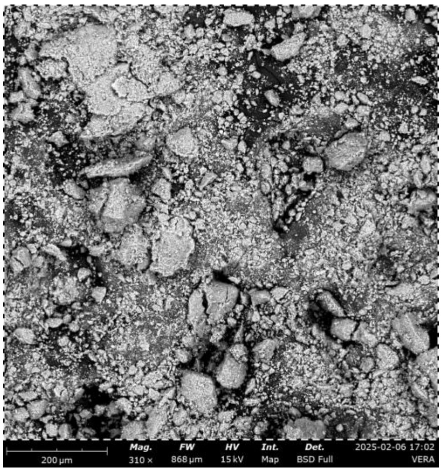

Figure 1: Clay SEM Image

One popular technique for describing the morphology of membranes is scanning electron microscopy, or SEM. It uses a concentrated electron beam to scan the sample's surface, transforms the signal it receives into grayscale data, and then shows it on the screen. The size of the beam spot at the sample surface is represented by each of the lines of image points that make up the SEM image. The diameter of this spot size (probe size) limits the SEM's capacity to resolve fine structures. Additionally, it is constrained by the probe's electron count [13].

Table 1: Optimization Results

RUN	Clay (g)	Ash (g)	NaOH (g)	Starch (g)	Apparent Viscosity (mPa.s)	Plastic Viscosity (mPa.s)
1	12.5	3	2	5	121.35	98.47
2	20	1.5	1.5	2	87.83	63.49
3	12.5	0	1.5	8	111.04	44.91
4	20	0	1.5	5	129.69	58.49
5	12.5	1.5	1.5	5	167.05	89.82
6	12.5	3	1.5	2	97.58	62.15
7	12.5	1.5	2	8	105.33	51.43
8	12.5	3	1	5	111.35	63.81
9	20	1.5	1	5	125.65	70.74
10	12.5	1.5	1	8	125.38	55.05
11	12.5	0	1	5	90.45	63.32
12	20	1.5	1.5	8	142.83	59.84
13	5	1.5	1.5	8	98.34	42.17
14	20	3	1.5	5	126.38	82.24
15	12.5	1.5	2	2	88.31	43.91
16	12.5	1.5	1.5	5	167.05	87.57
17	12.5	1.5	1	2	76.49	43.24
18	12.5	1.5	1.5	5	147.05	89.67
19	5	0	1.5	5	79.93	41.15
20	5	1.5	2	5	91.28	56.07
21	12.5	0	1.5	2	85.72	47.82
22	12.5	1.5	1.5	5	167.05	93.41
23	12.5	1.5	1.5	5	167.05	93.41
24	12.5	0	2	5	108.62	52.03
25	20	1.5	2	5	119.82	78.16
26	5	1.5	1.5	2	84.46	42.45
27	12.5	3	1.5	8	134.73	73.22
28	5	1.5	1	5	76.48	51.95
29	5	3	1.5	5	114.63	59.83

An ideal apparent viscosity of 116.431 mPa.s and a plastic viscosity of 75.6167 mPa.s were achieved using the following weights of the addictive's: 20g for clay, 1.5g for ash, 2g for NaOH, and 5g for starch [3]. The model's strong significance was revealed by the apparent viscosity analysis of variance (ANOVA) (F-value = 36.97, $p < 0.0001$). P-values for clay (A), ash (B), and starch (D) were less than 0.05, making them statistically significant among the four components [14].

Table 2: ANOVA for Apparent Viscosity

Source	Sum of Squares	df	Mean Square	F-value	p-value	
Model	22392.38	14	1599.46	36.97	< 0.0001	Significant
A-Clay	2916.58	1	2916.58	67.41	< 0.0001	
B-Ash	842.86	1	842.86	19.48	0.0006	
C-NaOH	69.65	1	69.65	1.61	0.2252	
D-Starch	3242.63	1	3242.63	74.94	< 0.0001	
AB	361.19	1	361.19	8.35	0.0119	
AC	106.40	1	106.40	2.46	0.1392	
AD	422.71	1	422.71	9.77	0.0074	
BC	16.69	1	16.69	0.3857	0.5446	
BD	34.99	1	34.99	0.8086	0.3837	
CD	253.92	1	253.92	5.87	0.0296	
A^2	4880.32	1	4880.32	112.79	< 0.0001	
B^2	3479.51	1	3479.51	80.42	< 0.0001	
C^2	6654.97	1	6654.97	153.81	< 0.0001	

D^2	6783.88	1	6783.88	156.79	< 0.0001	
Residual	605.75	14	43.27			
Lack of Fit	285.75	10	28.58	0.3572	0.9149	not significant
Pure Error	320.00	4	80.00			
Cor Total	22998.13	28				

The final coded equation for apparent viscosity shows that clay and starch contribute positively, while interaction terms like AB and CD negatively influence viscosity, indicating that increasing the clay and starch content improves viscosity, which is crucial for maintaining drilling fluid performance[15][6]. The adequacy of precision value (19.2750) further confirms that the model has an excellent signal-to-noise ratio for making reliable predictions. In addition, interaction effects like AB (clay-ash) and AD (clay-starch) also had significant contributions, and the quadratic terms A^2, B^2, C^2, and D^2 were all highly significant, indicating strong nonlinear relationships [10].

Table 3: Fit Statistics

R^2	**0.9737**
Adjusted R^2	0.9473
Predicted R^2	0.9067
Adeq Precision	19.2750
Std. Dev.	6.58
Mean	115.48
C.V. %	5.70

The Predicted R^2 of 0.9067 is in reasonable agreement with the Adjusted R^2 of 0.9473; i.e. the difference is less than 0.2. Adeq Precision measures the signal to noise ratio. A ratio greater than 4 is desirable. The ratio of 19.275 indicates an adequate signal. This model can be used to navigate the design space.

$$Apparent\ Viscosity = 163.05 + 15.59A + 8.38B + 2.41C + 16.44D - 9.50AB - 5.16AC + 10.28AD - 2.04BC + 2.96BD - 7.97CD - 27.43A^2 - 23.16B^2 - 32.03C^2 - 32.34D^2$$

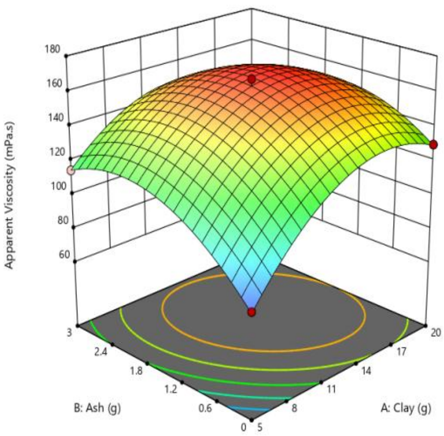

Figure 2: Relationship between Plastic Viscosity, Ash, and Clay

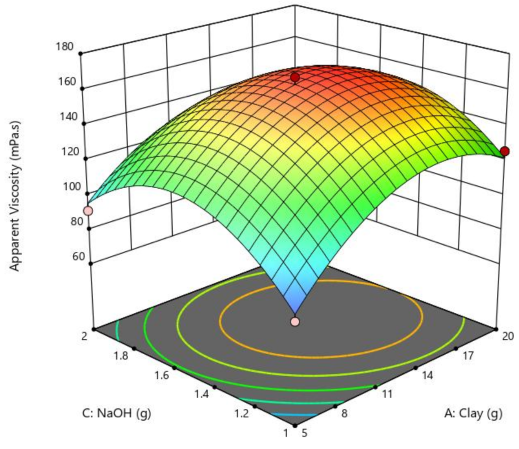

Figure 3: Relationship between Plastic Viscosity, NaOH, and Clay

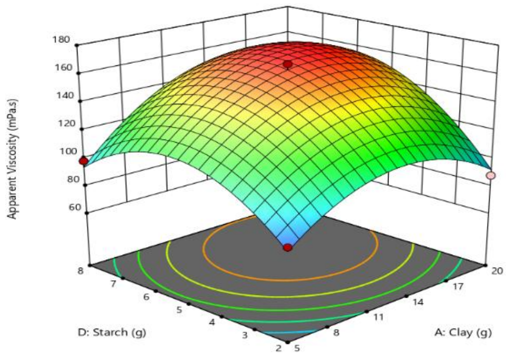

Figure 4: Relationship between Plastic Viscosity, Starch, and Ash

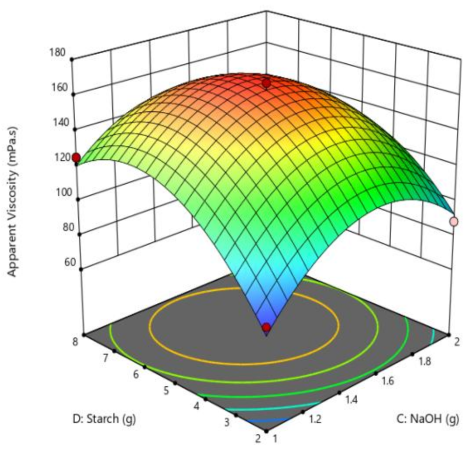

Figure 5: Relationship between Plastic Viscosity, Starch, NaOH

Table 4: ANOVA for Apparent Viscosity

Source	Sum of Squares	df	Mean Square	F-value	p-value	
Model	8652.29	14	618.02	22.03	< 0.0001	Significant
A-Clay	1186.84	1	1186.84	42.31	< 0.0001	
B-Ash	1452.00	1	1452.00	51.77	< 0.0001	
C-NaOH	85.12	1	85.12	3.03	0.1034	
D-Starch	46.26	1	46.26	1.65	0.2199	
AB	6.43	1	6.43	0.2291	0.6396	
AC	2.72	1	2.72	0.0971	0.7600	
AD	2.84	1	2.84	0.1012	0.7551	
BC	527.85	1	527.85	18.82	0.0007	
BD	48.86	1	48.86	1.74	0.2081	
CD	4.60	1	4.60	0.1640	0.6916	
A^2	1587.94	1	1587.94	56.61	< 0.0001	
B^2	720.43	1	720.43	25.69	0.0002	
C^2	1087.18	1	1087.18	38.76	< 0.0001	
D^2	4138.43	1	4138.43	147.55	< 0.0001	
Residual	392.68	14	28.05			
Lack of Fit	366.38	10	36.64	5.57	0.0561	not significant
Pure Error	26.29	4	6.57			
Cor Total	9044.97	28				

The Model F-value of 22.03 implies the model is significant. There is only a 0.01% chance that an F-value this large could occur due to noise. P-values less than 0.0500 indicate model terms are significant. In this case A, B, BC, A^2, B^2, C^2, D^2 are significant model terms. Values greater than 0.1000 indicate the model terms are not significant. If there are many insignificant model terms (not counting those required to support hierarchy), model reduction may improve your model. The Lack of Fit F-value of 5.57 implies there is a 5.61% chance that a Lack of Fit F-value this large could occur due to noise. Lak of fit is bad -- we want the model to fit. This relatively low probability (<10%) is troubling.

Table 5: Fit Statistics

R^2	**0.9566**
Adjusted R^2	0.9132
Predicted R^2	0.7621
Adeq Precision	14.5243
Std. Dev.	5.30
Mean	64.13
C.V. %	8.26

The Predicted R^2 of 0.7621 is in reasonable agreement with the Adjusted R^2 of 0.9132; i.e. the difference is less than 0.2. Adeq Precision measures the signal to noise ratio. A ratio greater than 4 is desirable. Your ratio of 14.524 indicates an adequate signal. This model can be used to navigate the design space.

$$Plastic\ Viscosity = 90.78 + 9.94A + 11.00B + 2.66C + 1.96D - 1.27AB - 0.825AC + 0.8425AD + 11.49BC + 3.5BD - 1.07CD - 15.65A^2 - 10.546B^2 - 12.95C^2 - 25.26D^2$$

3. Conclusion

The optimization study achieved an apparent viscosity of 116.431 mPa.s and a plastic viscosity of 75.6167 mPa.s using clay (20g), ash (1.5g), NaOH (2g), and starch (5g). ANOVA results confirmed the model's high significance, with good fits for both apparent and plastic viscosity. Clay and starch were identified as key factors for apparent viscosity, while clay and ash mainly influenced plastic viscosity, highlighting the potential for improved drilling mud performance through precise parameter adjustment.

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